

Visual Question Answering

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Introduction

Visual Question Answering (VQA) models combine visual and language understanding to answer questions about images. While their performance has improved significantly, it remains unclear whether these models actually focus on the parts of an image that a human would consider relevant when forming an answer.

In this study, we compare two distinct types of VQA models: the classification-based **ViLT** and the generative **BLIP** with a focus on answer attribution. We use attention visualizations to assess whether model attention aligns with human intuition depending on the question type.

Hypothesis

Generative models produce better results, but both attention maps align with human intuition.

Dataset description

- We use a subset of the **sajjadrauf/VQA** dataset.
- reformatted version of the popular VQAv2 benchmark
 - images paired with natural language questions
 - no ground-truth answers

Methodology

- We extract and visualize attention maps by overlaying the attention mask on the original image using a heatmap.
- To analyze model behavior, we generate two types of visualizations:
- **Token-wise attention maps** to show the spatial focus of the model for each individual question token.
 - **Averaged attention maps** to highlight overall focus by combining attention across all tokens.

Example dataset entry



ViLT - example averaged map



BLIP - example averaged map



Conclusion

The answers generated by BLIP were of higher quality (see example), so the first part of the hypothesis turned out to be true. However, attention maps did not highlight the most important features, so the second part of the hypothesis was not supported by the results.

References

ViLT model on Hugging Face: <https://huggingface.co/dandelin/vilt-b32-finetuned-vqa>
ViLT paper: <https://arxiv.org/abs/2102.03334>
BLIP model on Hugging Face: <https://huggingface.co/Salesforce/blip-vqa-base>
BLIP paper: <https://arxiv.org/abs/2201.12086>
VQA Dataset website: <https://visualqa.org/>
Dataset: <https://huggingface.co/datasets/sajjadrauf/VQA>